

Proposed Solution

Introduction

Diabetic Retinopathy is a serious eye disease that occurs due to long-term diabetes and can lead to blindness if not detected early. Early diagnosis plays a crucial role in preventing severe vision loss. The proposed solution aims to develop an automated system that uses deep learning techniques to detect diabetic retinopathy from retinal images.

Objective

The main objective of this project is to design and implement a deep learning-based system that can automatically analyze retinal images and identify the presence of diabetic retinopathy. The system will help doctors and healthcare professionals in early diagnosis and decision-making.

Proposed System

The proposed system uses a Convolutional Neural Network (CNN) model to analyze retinal images. The CNN model is trained using a dataset of labeled retinal images. The system learns patterns and features associated with diabetic retinopathy and uses them to classify new images.

The process of the proposed system includes the following steps:

Image Input: The user uploads a retinal image to the system.

Preprocessing: The image is resized and normalized to make it suitable for the deep learning model.

Feature Extraction: The CNN automatically extracts important features from the retinal image.

Classification: The trained model predicts whether the retinal image shows signs of diabetic retinopathy.

Result Display: The system displays the prediction result to the user.

Technologies Used

- The proposed system uses the following technologies:
- Programming Language: Python
- Deep Learning Framework: TensorFlow / Keras
- Image Processing Library: OpenCV
- Development Environment: Jupyter Notebook or Python IDE

Advantages of the Proposed System

- Faster detection of diabetic retinopathy
- Reduced workload for doctors
- High accuracy in medical image classification
- Early diagnosis leading to better treatment

Conclusion

The proposed deep learning-based system will assist healthcare professionals in detecting diabetic retinopathy efficiently. By using automated image analysis, the system can improve early diagnosis and help prevent vision loss in diabetic patients.